Earth and Planetary Science Letters 309 (2011) 302–317

Contents lists available at ScienceDirect

Earth and Planetary Science Letters

journal homepage: www.elsevier.com/locate/epsl

Revisiting the magnetostratigraphy of the Central Atlantic Magmatic Province (CAMP) in Morocco

E. Font [a,*], N. Youbi [b,c,d], S. Fernandes [a], H. El Hachimi [b], Z. Kratinová [e], Y. Hamim [b]

[a] IDL-UL, Instituto Dom Luís, Universidade de Lisboa, Portugal
[b] Department of Geology, Cadi Ayyad University, Marrakech, Morocco
[c] National Centre for Scientific and Technical Research, Rabat, Morocco
[d] Centro de Geologia da Universidade de Lisboa (CeGUL), Portugal
[e] Institute of Geophysics of the Academy of Sciences of the Czech Republic, Prague, Czech Republic

ARTICLE INFO

Article history:
Received 19 April 2011
Received in revised form 28 June 2011
Accepted 10 July 2011
Available online 10 August 2011

Editor: P. DeMenocal

Keywords:
CAMP
Morocco
remagnetization
magnetostratigraphy
Triassic–Jurassic boundary

ABSTRACT

The origin of the Triassic–Jurassic (Tr–J) mass extinction is still a matter of debate: proponents of the idea that continental flood basalts of the Central Atlantic Magmatic Province (CAMP) are responsible for the crisis are opposed by those who favor an extraterrestrial origin linked to the impact of meteorite. Principal limitations reside in the difficulty to date and correlate CAMP lavas with the marine realm turnover. One argument widely used to suggest that CAMP lavas pre-dated the Tr–J boundary in Morocco is based on the presence of two brief magnetic reversals in the intermediate units of the Tiourjdal and Oued Lahr sections (Morocco) that were correlated to the E23r chron from the Newark basin and to the SA5n.2r/3r and SA5r chrons of the Saint Audrie Bay [Knight, K.B., Nomade, S., Renne, P.R., Marzoli, A., Betrand, H., Youbi, N., 2004. The Central Atlantic Magmatic Province at the Triassic–Jurassic boundary: paleomagnetic and $^{40}Ar/^{30}Ar$ evidence from Morocco for brief, episodic volcanism. Earth and Planetary Science Letters 228, 143–160]. However the primary origin for these negative (reverse) magnetic components is questionable since no field or reversal test was provided to constrain the primary character of the remanence as well as because the small number of samples. Here we have conducted a detailed paleomagnetic and magnetic mineralogy study of the interbedded limestones of the Tiourjdal section and of other CAMP lavas sections where the intermediate unit is complete, namely the Tizi El Hajaj, Jbel Imzar and Aït Ourir sections, to better constrain the origin and stratigraphic location of these negative magnetic components. We show that the interbedded limestones of the Tiourjdal section were entirely remagnetized by chemical processes via acid and oxidizing hydrothermal fluids generated by eruptions of CAMP lavas. In addition, magnetostratigraphic data of the Tizi El Hajaj, Jbel Imzar and Aït Ourir sections show that the entire intermediate unit encompassed a positive (normal) magnetic interval. A good quality paleomagnetic pole for the CAMP lava in Morocco is then provided (Plat = 60.0°; Plong = 241.6°; A95 = 2.6; N = 99) that is now in better agreement with its trans-Atlantic counterpart.

© 2011 Elsevier B.V. All rights reserved.

1. Introduction

The Triassic–Jurassic event is one of the most severe mass extinctions of the Phanerozoic, the so-called "big five" (e.g., Newell, 1967; Raup and Sepkoski, 1982), where more than 50% of marine and continental species disappeared, opening the way to the ascent of dinosaurs (Olsen et al., 2002a,b). During the last decades, there has been a vigorous debate about the stratigraphic position of the Triassic–Jurassic (Tr–J) boundary and the mechanisms that triggered the associated mass extinction. Proponents of the idea that continental flood basalts of the Central Atlantic Magmatic Province (CAMP) are responsible for the Tr–J crisis (e.g., Courtillot and Renne, 2003; Knight et al., 2004; Marzoli et al., 2004; Schaltegger et al., 2008; Verati et al., 2007) are opposed by those who favor an extraterrestrial origin linked to the impact of meteorite (Olsen et al., 1987; Ward et al., 2001; Olsen et al., 2002a). Alternatively, some authors suggest that iridium anomalies can also be originated by mantle outgassing and aerosol deposition during early stages of the CAMP lavas eruption (e.g., Morgan et al., 2004; Tanner et al., 2008). Principal limitations reside in our difficulty to date and correlate the onset of the CAMP in continental areas to the marine realm turnover (e.g., Deenen et al., 2010; Lucas et al., 2011; Lucas and Tanner, 2007; Marzoli et al., 2004, 2008; Whiteside et al., 2007, 2008). Indeed, classical radiometric methods are inadequate to constrain the timing and duration of the CAMP because of errors due to inter-laboratory and standard calibration (e.g. Kuiper et al., 2008; Renne et al., 2010) leaving paleomagnetic techniques as the most suitable alternative to provide reliable geochronological constrains for the CAMP lavas.

* Corresponding author at: IDL-UL, Instituto Dom Luiz, Universidade de Lisboa, Edifício C8, Campo Grande, 1749-016, Lisboa, Portugal. Tel.: +351 217500811.
E-mail address: font_eric@hotmail.com (E. Font).

0012-821X/$ – see front matter © 2011 Elsevier B.V. All rights reserved.
doi:10.1016/j.epsl.2011.07.007

In Morocco, two brief reversals were identified in the intermediate unit of the CAMP lavas (Tiourjdal and Oued Lahr) section (Knight et al., 2004; Marzoli et al., 2004) that the authors correlated with the SA5n.2r and SA5n.3r intervals of the St. Audrie's Bay section (Hounslow et al., 2004) and the chron E23r of the Newark basin (Kent and Olsen, 1999) thus suggesting that CAMP volcanism predated the Tr–J boundary. Conversely, Whiteside et al. (2007, 2008) rather correlated these reversals with the early Jurassic polarity reversals in the Paris basin Montcornet core (Yang et al., 1996). More recently, Deenen et al. (2010) found an interval of negative magnetic polarity within the infrabasaltic red siltites of the Argana Basin, Morocco, that the authors correlated with the E23r of the Newark Basin. On this basis, Deenen et al. (2010) correlated the reversals found by Knight et al. (2004) to the SA5r interval of the Saint Audrie's Bay (Hounslow et al., 2004). These contrasting scenarios clearly highlight the need to provide better constraints on the origin and stratigraphic location of the magnetic reversals found in the CAMP lavas. Furthermore, the primary origin (i.e. remanence close in time to cooling phase or to sediment deposition/precipitation) of these magnetic reversals must be strongly questioned for several reasons: i) absence of any field or statistical tests that constrain the primary character of the magnetic remanence; ii) small number of samples for the reverse magnetic component; and iii) non-repeatability of the stratigraphic position of these reversals within other contemporaneous lava flows. In addition, one reversal (i.e. Tiourjdal section) was found in thin (<2 m) interbedded limestones, rocks which are known to be easily prone to remagnetization (e.g., Font et al., 2006; Jackson, 1990; Jackson et al., 1992, 1993; McCabe and Elmore, 1989; Weil and Van der Voo, 2002; Xu et al., 1998).

To unravel the primary vs. secondary origin of the magnetic reversal found in CAMP lavas from Morocco, we present here a detailed paleomagnetic and rock magnetic study of the interstratified limestones of the Tiourjdal section previously studied by Knight et al. (2004). To check for the repeatability of the second reversal, located at the base of the intermediate unit at Oued Lahr, we also sampled other complete intermediate units, namely the Tizi El Hajaj, Jbel Imzar and Aït Ourir sections.

2. Geological context and sampling

The Central Atlantic Magmatic Province (CAMP; Fig. 1) was emplaced in circum-Atlantic basins in Europe, Africa, North and South America, during the rifting of the supercontinent Pangea (e.g. Marzoli et al., 1999, 2006). It is mainly composed of low-Ti tholeiitic basalts that erupted over a total surface in excess of 10^7 km^2 and a total magma volume estimated at $2–4\times10^6$ km^3, making this one of the largest known Phanerozoic igneous provinces (e.g. Marzoli et al., 2011). The best preserved and most complete CAMP lava sequences are well exposed in the High Atlas Mountains in Morocco. It was divided into four primary formations called the lower, intermediate, upper and recurrent units (Bertrand et al., 1982; Marzoli et al., 2004; Youbi et al., 2003). They are separated by thin sedimentary units (siltstones, sandstones, stromatolitic limestones) and paleosols that represent minor periods of volcanic quiescence.

For this study we basically followed the trail of Knight et al.'s (2004) sampling of the Tiourjdal and Aït Ourir sections, with the exception of the Tizi El Hajaj section and the Intermediate Unit of the Jbel Imzar section which were never studied until now. We essentially focused our sampling on the lavas from the intermediate unit where the magnetic reversals have been found, except for the Tizi El Hajaj section where the Lower Unit was also sampled. For the distinction of the different lava flow units of the CAMP volcanic successions, we followed the terminology of Self et al. (1997, 1998).

The Tiourjdal section crops out in the Central part of the High Atlas Mountains and is located between the city of Marrakech and Ouarzazate. It is considered as one of the most complete CAMP volcanic successions in Morocco and consists of massive and well preserved pahoehoe basaltic flows gently dipping ~15° SSW and oriented N~125°. The interbedded limestones where the reversal was found (Knight et al., 2004) are located in the middle part of the Intermediary Unit. Here, more than 19 paleomagnetic sites (~60 samples) were collected along the 60 cm of the interbedded limestones (Fig. 2).

The Tizi El Hajaj section is found in the Argana basin and crops out 30 km to the northeast of Agadir city in the Western High Atlas (Fig. 1). It is one of the most complete CAMP volcanic successions of the basin (Aït Chayeb et al., 1998; El Hachimi et al., 2011). It consists of massive and well preserved pahoehoe basaltic flows gently dipping 30° to the NNW and N245° oriented. The Lower and Intermediary Units are separated by a 0.5–3 m thick silty/sandy layer (Aït Chayeb et al., 1998). We collected a total of 64 samples in both lavas core and lavas crust from the Lower and Intermediate Unit.

The Jbel Imzar section crops out in the Central High Atlas, east of the city of Marrakech. It consists of pahoehoe basaltic flows oriented N230° with dip of 38° to the NW (Fig. 1). The base of the Intermediary unit is marked by a thin (<1 m) and discontinuous siltite layer overlain by pillow lavas. We collected 61 samples in the Intermediate Unit that has never been studied yet (Knight et al., 2004).

The Ait Ourir section crops out in the Atlas system, east of the city of Marrakech. Basaltic flows are oriented N30° with dip of 30° SE (Fig. 1). The base of the intermediate Unit is represented by a thin siltite layer overlain by pillow while the top is marked by the occurrence of paleosoils. We collected 38 samples in the Intermediate Unit.

3. Methods

Samples were treated by Alternating Field (AF) through several progressive and narrow steps using a LDA-3A demagnetizer. Remanence measurements were performed using a three-axis 2G-cryogenic magnetometer (for limestones) and a JR6 magnetometer (for lavas). Characteristic Remanent Magnetizations (ChRM) were calculated using Principal Component Analysis (Kirschvink, 1980) and Fisher (1953) statistics. Mean magnetic components, Virtual Geomagnetic Poles (VGPs) and cut-off angles (Vandamme, 1994) were calculated using the Rotpole software.

IRM and ARM was acquired in applied fields up to 100 mT using an impulse magnetizer IM-10-30 and samples were subsequently measured with a JR6 magnetometer. ARM was acquired under an AF field of 100 mT biased with a DC field of 0.1 mT using a LDA-3A demagnetizer coupled with AMU-1A anhysteretic magnetizer. The frequency dependence of the magnetic susceptibility was obtained with a Bartington MS2 dual-frequency (470 and 4700 Hz) alternating current bridge on whole rock specimens. Hysteresis and remanent parameters were acquired by a Vibrating Sample Magnetometer (Model EV9 VSM) at the Institute of Geophysics in Prague. SEM observations and EDS analyses were performed on carbon-coated rock fragments cut from the paleomagnetic samples using a Jeol JSM-6360LV coupled to a Noran Instrument EDS analyzer.

4. Limestones of the Tiourjdal section

4.1. Paleomagnetism

Samples carry a weak natural remanent magnetization (NRM) varying from 10^{-2} to 10^{-4} A/m (Fig. 2; Table X1). From the 41 analyzed samples, most of the remanence is cleaned below 30 mT indicating a very low coercive phase as principal magnetic carrier. Orthogonal plots show uni- and bimodal directions that rapidly scattered after demagnetization at 10 mT. The most stable remanent magnetization was isolated between 2 and 20 mT, after which it reaches the noise level our 2G-cryogenic (Fig. 2). Sample-based

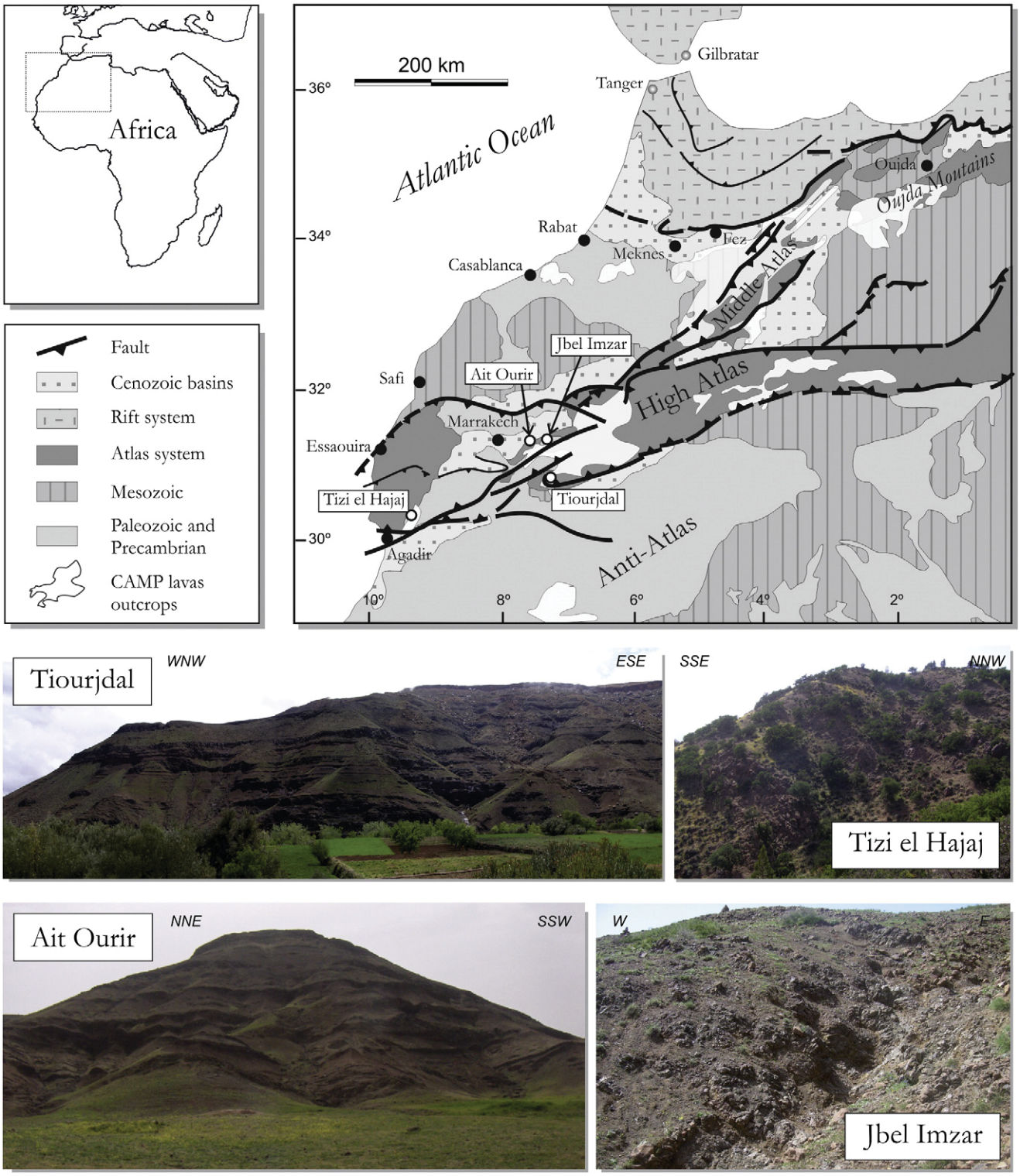


Fig. 1. Geological map of northern Morocco (modified after Hafid et al., 2006) and field photographs of the studied sections: Tiourjdal (N31°07′29″; W7°22′48″), Ait Ourir (N31°32′40″; W7°40′26″), Tizi el Hajaj (N31°32′40″; W7°25′50″) and Jbel Imzar (N30°38′59″; W9°17′44″) sections.

means of the low coercive remanence are positive (northward directed) oriented approximately N340° with considerably scattered inclinations. No geomagnetic reversals were found in our samples. These magnetic features rather suggest a secondary magnetic overprint for these rocks, probably corresponding to a chemical or thermo-viscous remanent magnetization overprint.

4.2. Magnetic mineralogy

4.2.1. Frequency-dependent magnetic susceptibility and ARM/SIRM

The frequency dependence of the magnetic susceptibility allows evaluating the contribution of ultrafine (superparamagnetic) particles to the bulk magnetic properties. Basically, at high frequency

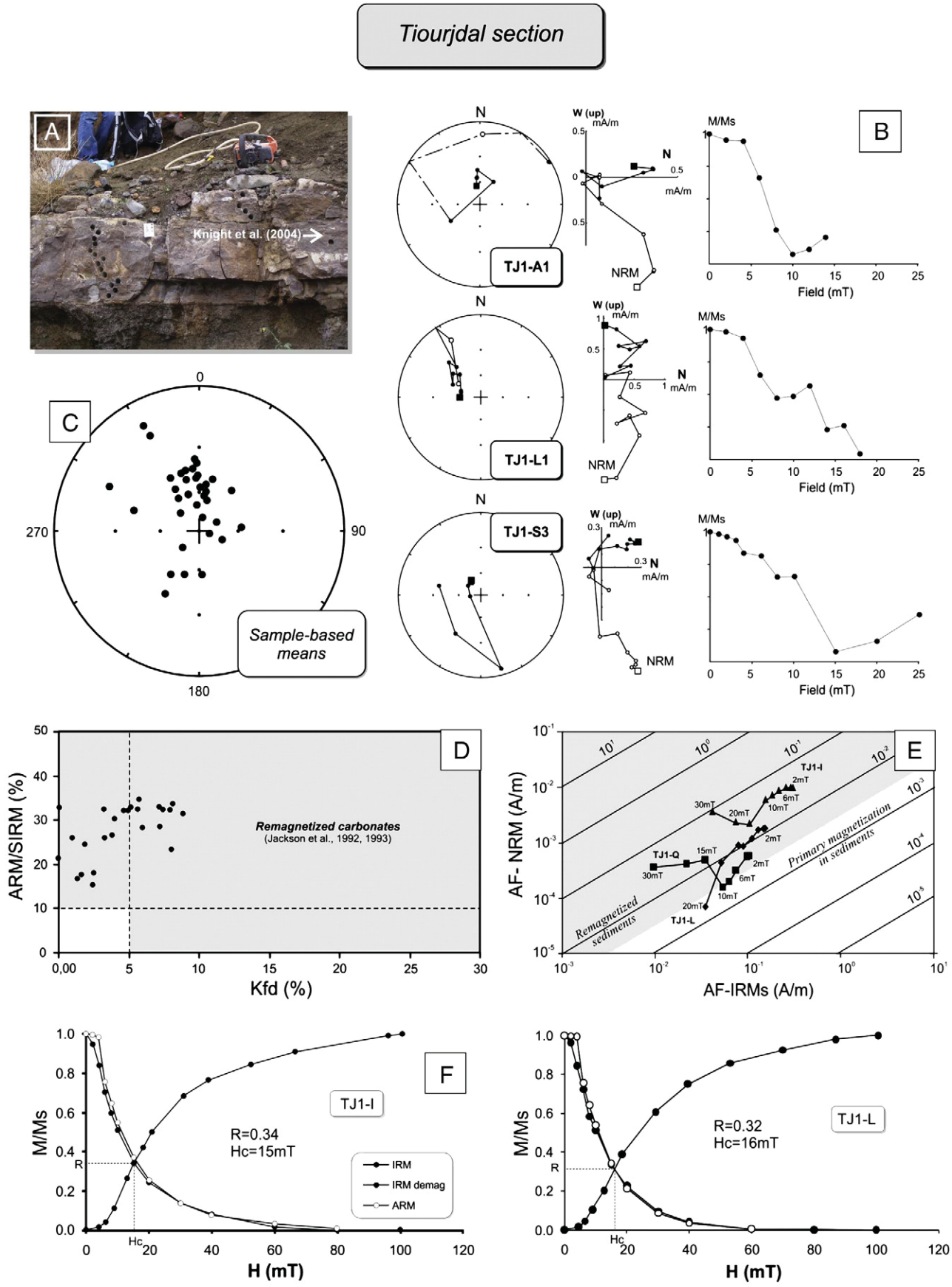


Fig. 2. Rock magnetic properties of the interbedded limestones of the Tiourjdal section. A) Field photograph showing our detailed sampling and location of one Knight et al.'s (2004) sample; B) Stereographic and orthogonal projections and intensity remanence vs. AF demagnetization showing that very low coercive phases are the principal magnetic carriers in this rocks; C) Sample-based mean directions showing that all samples carry a positive (normal) remanent magnetization; D) Frequency-dependent magnetic susceptibility (Kfd) and ARM/SIRM ratios typical of remagnetized carbonates; E) AF demagnetization NRM:IRM ratio typical of remagnetized sediments (Fuller et al., 2002); and F) results of the Lowrie–Fuller (Johnson et al., 1975) test.

(4700 Hz) the measurement time is short enough for these particles to behave as stable SD grains while at low frequency (470 Hz) the measurement time is longer than the relaxation time and particles behave as superparamagnetic (SP). In the case of carbonates it represents a powerful magnetic proxy to check for the occurrence of remagnetization processes (Font et al., 2005, 2006; Jackson et al., 1993; Jackson and Worm, 2001; Zwing et al., 2005). Indeed, remagnetized carbonates are characterized by values of Kfd ≥ 5% (Kfd = [Khf − Klf] / Klf where Khf is high frequency susceptibility and Klf is low frequency susceptibility). Remagnetized carbonates can also be detected by ARM/SIRM ratio that is greater than 10% (Jackson et al., 1992). In our samples, Kfd values and ARM/SIRM ratios are in the typical ranges of remagnetized carbonates (Fig. 2).

4.2.2. Lowrie–Fuller test

The modified Lowrie–Fuller test (Johnson et al., 1975) was originally designed to distinguish between single-domain (SD) and multi-domain (MD) magnetite grains, the ARM being harder than the IRM in SD grains. The test was applied to three samples (TJ1-I, TJ1-L, TJ1-Q) for which results are illustrated in Fig. 4. As maximum AF demagnetization field does not exceed 100 mT in our equipment, only size of the lower coercive phases can be assessed. All samples show similar IRM and ARM demagnetization pattern above AF demagnetization fields at 6 mT (Fig. 2) suggesting a mixture of single and multidomain particles. Below AF demagnetization fields at 6 mT, ARM demagnetization curves exhibit a characteristic plateau interpreted to result from the presence of SD particles.

The Cisowski (1981) test consists in comparing the AF and IRM acquisition coercivity spectra to detect interactions, either between single domain magnetite grains or within MD grains. Results give an R value (interaction degree) between 0.30 and 0.34 indicating low to moderate contributions from MD ferromagnetic particles (Fig. 2). Approximation of the remanence coercive force (Hcr) given by the projection of R on the abscissa axis is between 15 and 17 mT indicating that very low coercive minerals are the dominant magnetic carriers.

4.2.3. NRMs:IRMs

The comparison of the magnitude and demagnetization patterns of the NRM with those of saturation IRM provide a good means to distinguish primary from secondary magnetizations in magmatic and sedimentary rocks (Fuller et al., 2002). Sedimentary rocks carrying a primary (detrital) remanent magnetization yield NRM:IRMs ratios of parts in 1000, whereas remagnetized sediments give ratios of parts in 100. In addition, remagnetized sedimentary rocks are characterized by concave up curves that reflect the admixture of hard material in samples carrying secondary magnetization. Here, we calculated the NRM:IRMs ratios of the same samples processed by the Lowrie–Fuller and Cisowki test (i.e., TJ1-I, TJ1-L, TJ1-Q) for which results are illustrated in Fig. 2. All samples show NRM:IRMs ratios in the range of 10^{-2} similar to those of remagnetized sediments. Samples TJ1-I and TJ1-Q show typical concave up curves that, in corroboration with previous data, confirm the admixture of hard material.

4.2.4. Hysteresis loops and remanent parameters

Fourteen hysteresis curves were measured from −20,000 Oe to 20,000 Oe on crushed samples in order to characterize the basic magnetic parameters and estimate the grain-size distribution. The linear slope subtraction (from 10,000 Oe) was applied to correct the diamagnetic or paramagnetic influence. The samples in general exhibit a weak signal with small remanence (average Mrs = 8e −5 emu/g) and coercivity (Hc ~ 65 Oe). The majority of hysteresis loops (Fig. 3A) are wide open up to relatively high magnetic field Some of the hysteresis curves are wide open only at low field suggesting a presence of SD particles.

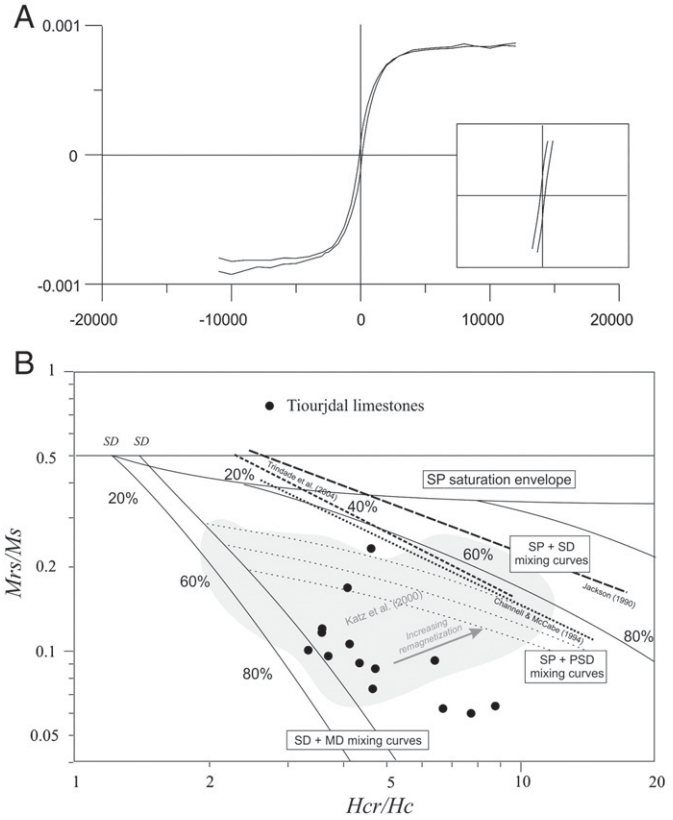


Fig. 3. A) Example of the hysteresis loop. The inset shows the wasp-waisted shape of the hysteresis at low fields. B) Day plot showing the domain state of samples compared with the theoretical mixing curves after Dunlop (2002a,b) and hysteresis data from remagnetized carbonates (Jackson, 1990; Channell and McCabe, 1994; Katz et al., 2000; Trindade et al., 2004).

The ratio of saturation remanence to saturation magnetization, Mrs/Ms, against the ratio of remanent coercive force to ordinary coercive force, Hcr/Hc, the so called Day plot (Day et al., 1977; Dunlop, 2002a,b) was constructed to characterize the domain state of magnetic particles. The data falls (Fig. 3B) in region in between the SD + MD and the SP + SD model curves (Dunlop, 2002b). The majority clusters close to the SD + MD mixing curve around $f_{MD} \sim 75\%$. However, there is a large dispersion towards the SD + 10-nm SP mixing curve which can be interpreted in terms of binary mixtures of PSD and SP grains (dot–dashed curves) (Dunlop, 2002b).

4.2.5. Microscopic observations

Scanning Electron Microscopy (SEM) coupled with Energy Dispersive Spectra (EDS) is a powerful tool to identify the nature and origin (primary or secondary) of the magnetic carriers. Detailed SEM-EDS analyses were done on samples TJ1-I, TJ1-L and TJ1-Q for which results are illustrated in Fig. 4. Sample TJ1-I contains abundant sulfates and sulfides such as calcium sulfate ($CaSO_4$, Fig. 4A) and pyrite (FeS_2) probably hosted from copper sulfate ($CuSO_4$, Fig. 4B and D). Concretions of Cu, Fe, Mn, Cl and Ca are locally found (Fig. 4C). Some very fine iron oxides are found coating clay minerals in sample TJ1-L (Fig. 4E). Transition metals such as Cu and Mo are observed filling fractures (Fig. 4F). Rare relics of titanomagnetite, severely oxidized, are found in sample TJ1-Q (Fig. 4G and H). All these evidences point to a second origin for the magnetic mineralogy of the interbedded Tiourjdal limestones probably from hydrothermal fluid circulations and enrichment in epigenetic sulfur and transition metals. Iron oxides (i.e. magnetite) are rare or too small to be observed under the microscope.

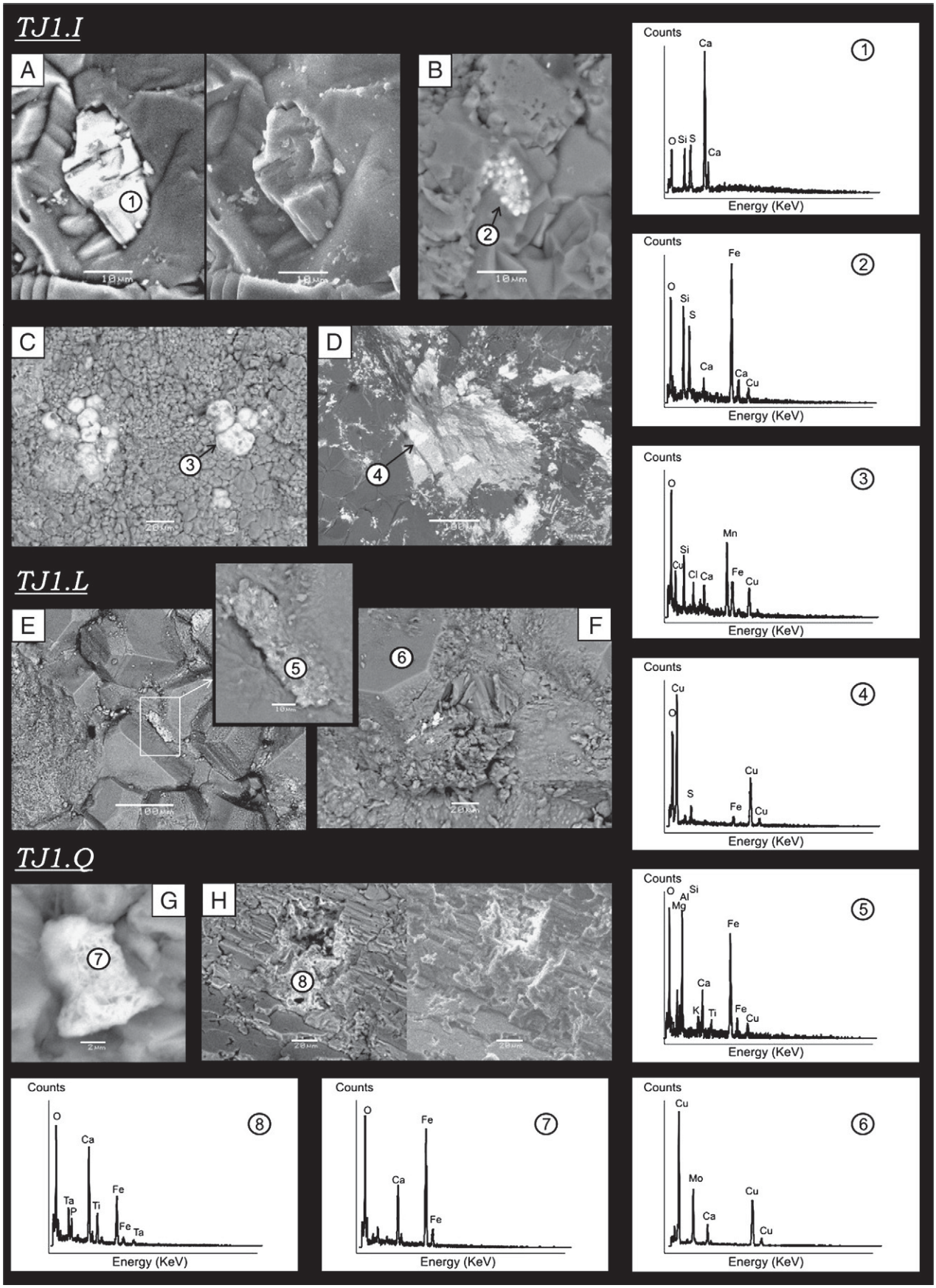


Fig. 4. Scanning Electron Microscopy (SEM) coupled with Energy Dispersive Spectra (EDS). See text for details.

5. Magnetostratigraphy of the intermediate unit of the camp lavas in Morocco

5.1. Tizi El Hajaj section

A total of 64 samples were collected in the Tizi El Hajaj (TH) section (47 in the Lower Unit and 17 in the Intermediate unit) for which 58 show stable AF demagnetization patterns, equivalent to a percentage of success of 91% (Table X2). After stepwise AF cleaning, most samples show stable demagnetization patterns where more than 90% of the remanence is cleaned up to 60 mT, except in rare samples (TH2-B2) where harder ferromagnetic mineral (i.e. hematite) may contribute to the bulk magnetic properties (Fig. 5). IRM analyses treated by the cumulative log-Gaussian function (Kruiver et al., 2001; Robertson and France, 1994) indicated the presence of a low to medium coercivity phase (Fig. 6). Curie temperature of 560–580 °C obtained by thermomagnetic analyses indicated a Ti-poor titanomagnetite as principal magnetic carrier (Fig. 6). Orthogonal plots show mostly univectorial distributions that include the origin, while a few samples show the contribution of a secondary, probably viscous in origin, remanence that is easily cleaned up to 4 mT (Fig. 5). Magnetic orientations are positive (normal) and N° to NNW° directed with mean inclinations of 20 to 25°. Only one lava flow (TH6), localized at the top of the inferior unit, gave unreliable and scattered sample-based mean directions (Fig. 8).

To check for the similarity of the mean ChRM from the Lower and Intermediate Unit we applied the test of McFadden and Lowes (1981) that discriminate mean directions, that follow a Fisher distribution, using the N (number of unit vectors), R (length vector) and K (dispersion) parameters. Results indicate that the Lower and Intermediate Unit do not share a common mean at 95% confidence. This is in agreement with previous data from the Tiourjdal section where the limit between the Lower (component DG2) and Intermediate (component DG3) Unit is marked by a significant change in declination and inclination values (Knight et al., 2004).

Finally we calculated the mean ChRM and corresponding mean paleomagnetic pole based on in-situ and tilt corrected sample-based mean directions for both the Lower and Intermediate Unit (Fig. 5). Tilt corrected results give a mean ChRM of D = 349.2°; I = 21.7° (N = 36, α_{95} = 2.1) and D = 335.3°; I = 23.9° (N = 19, α_{95} = 6.6) for the Lower and Intermediate Unit, respectively. Corresponding paleomagnetic poles are located at Long. = 200.5°; Lat. = 68.2° (A95 = 1.7°; Paleolat. = ± 11.2) and Long. = 227.6°; Lat. = 61.1° (A95 = 6.5°; Paleolat. = ± 12.5), respectively (Table 1).

5.2. Jbel Imzar section

A total of 61 samples were collected in the intermediate unit of the Jbel Imzar (JI) section for which 55 show stable AF demagnetization patterns, equivalent to a percentage of success of 90% (Table X2). Results are similar to those obtained in the basalts from the Tizi El Hajaj section. Samples show stable demagnetization patterns where more than 90% of the remanence is cleaned between 20 and 60 mT depending on samples (Fig. 7). Median destructive fields (MDF) are around 10–20 mT suggesting a low coercive (Ti-rich) magnetite as principal magnetic carriers. Rare samples show admixture of small amounts of hematite (JI17-3; Fig. 7). Orthogonal plots show mostly univectorial distributions including origin while a few samples show the contribution of a viscous remanent magnetization removed up to 4 mT (Fig. 7). Some AF demagnetization curves show a plateau until 6 mT (JI11-2; Fig. 7) that we interpreted to be related to the admixture of SD minerals to the bulk properties. Samples from site JI20 and JI21 were collected in pahoehoe lobes and toes that appeared strongly oxidized in the field thus explaining the presence of much lower coercive magnetite (MDF<10) as principal carriers as well as the very low remanence intensity observed in these samples (JI20-1; Fig. 7). Mean magnetic directions are positive (normal) and N° to NW° directed with mean inclinations of 20 to 25° (Fig. 7). Mean ChRM and corresponding mean paleomagnetic pole based on tilt corrected sample-based mean directions are D = 338.6°; I = 23.3° (N = 54, α_{95} = 3.5) and Long. = 238.2°; Lat. = 62.5° (A95 = 3.0°), respectively. Paleolatitude computed from the mean paleomagnetic pole is ± 12.1° (Table 1).

We applied the test of McFadden and Lowes (1981) to sample means of the TH (Intermediate Unit) and JI data sets. Results show that both sections share a common mean at 95% confidence (Table X3). Conversely, the test gives a negative response when comparing JI sample means with those from the Lower Unit of the Tizi el Hajaj section. This suggested that our tilt correction were correct and that lavas flows from the TH (Int.) and JI sections studied here are mostly synchronous within the Intermediate Unit.

5.3. Ait Ourir section

A total of 38 samples were collected in the intermediate unit of the Ait Ourir (AO) section for which 31 show stable AF demagnetization patterns, equivalent to a percentage of success of 82% (Table X2). Samples show stable demagnetization patterns but most of the remanence is rapidly cleaned at ~20 mT. Consequently, directions rapidly scattered at low AF demagnetization fields (AO18-1; Fig. 8). Unblocking temperatures of this low coercive phase are about 520 °C indicating that Ti-rich titanomagnetite is the principal magnetic carrier in these rocks (Fig. 8). Most samples show bi-vectorial magnetic directions in both stereographic projection and orthogonal plots suggesting that a viscous or thermoviscous magnetization overprinted the original, primary, remanence (Fig. 7). After tilt correction, all mean magnetic directions calculated for higher AF fields and origin are positive (normal) and N° to W° directed with mean inclinations of 30° (Fig. 8). Mean ChRM and corresponding mean paleomagnetic pole based on tilt corrected sample-based mean directions are D = 316.6°; I = 35.9° (N = 29, α_{95} = 5.0) and Long. = 260.6°; Lat. = 47.7° (A95 = 4.6°; Paleolat = ± 12.1°), respectively (Table 1). These values well agree with previous data from the same section (Knight et al., 2004) but are significantly different from those calculated for the TH and JI sections. Indeed, AO sample means failed the McFadden and Lowes (1981) test when compared to the TH and JI data sets. This discrepancy can be linked to error in tilt correction in the Ait Ourir section, where bedding orientation is difficult to discern in the field (Knight et al., 2004). Another possibility is to consider very rapid eruptions where secular variations of the Earth magnetic field have not been minimized. Indeed, the dispersion in sample-based mean directions is mostly observed along declination, from W° to N° directed, and this particular pattern is also present in the TH and JI section (Figs. 7 and 8). Because the time of the acquisition of a thermal remanent magnetization in basalt not only depend on the time of the eruption but also on relaxation times of the magnetic carriers of the rocks, which are significantly distinct in the Ait Ourir section, it can explain the discrepancy observed in the directions of the ChRM.

6. Discussion

6.1. Remagnetization of the interbedded limestones from the Tiourjdal section

From a paleomagnetic point of view, the reversals identified by Marzoli et al. (2004) and Knight et al. (2004) are questionable. Assuming that the Earth's magnetic field is similar to a Geocentric Axial Dipole (GAD), a geomagnetic field reversal corresponds to a 180° change in surface geomagnetic field direction at all points (Cox et al., 1964; Matuyama, 1929; Mercanton, 1926). In this perspective, McFadden and McElhinny (1990) elaborate a simple test based on

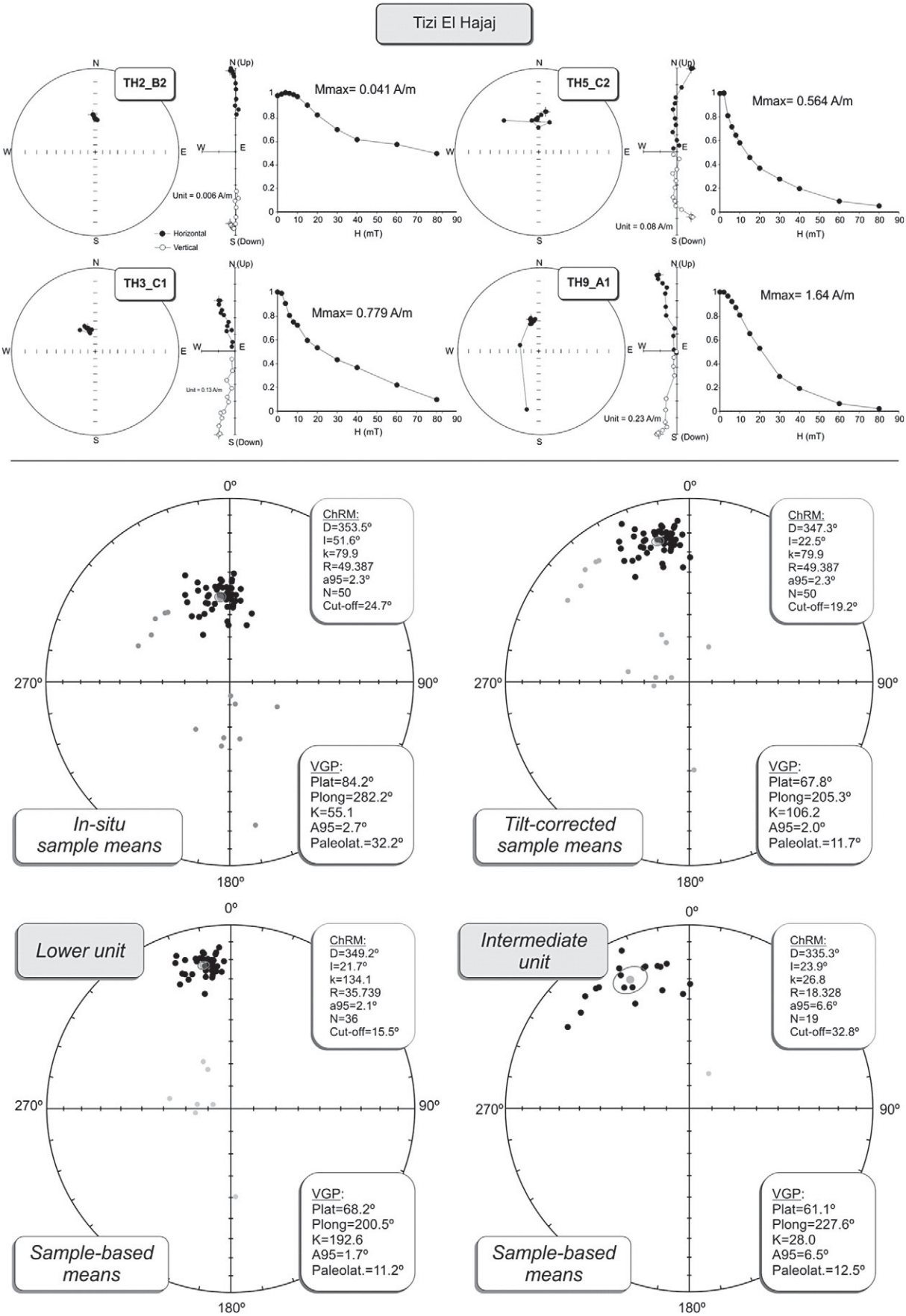


Fig. 5. Paleomagnetic results for the Tizi el Hajaj section. (Above) Stereographic and orthogonal projections and intensity remanence vs. AF demagnetization; (below) Sample-based mean directions and mean Characteristic Remanent Magnetization (ChRM) calculated using the Rotepole program and applying the Vandamme (1994) cut-off. Outliners are represented by gray data points. Corresponding Virtual Geomagnetic Poles (VGPs) are also indicated.

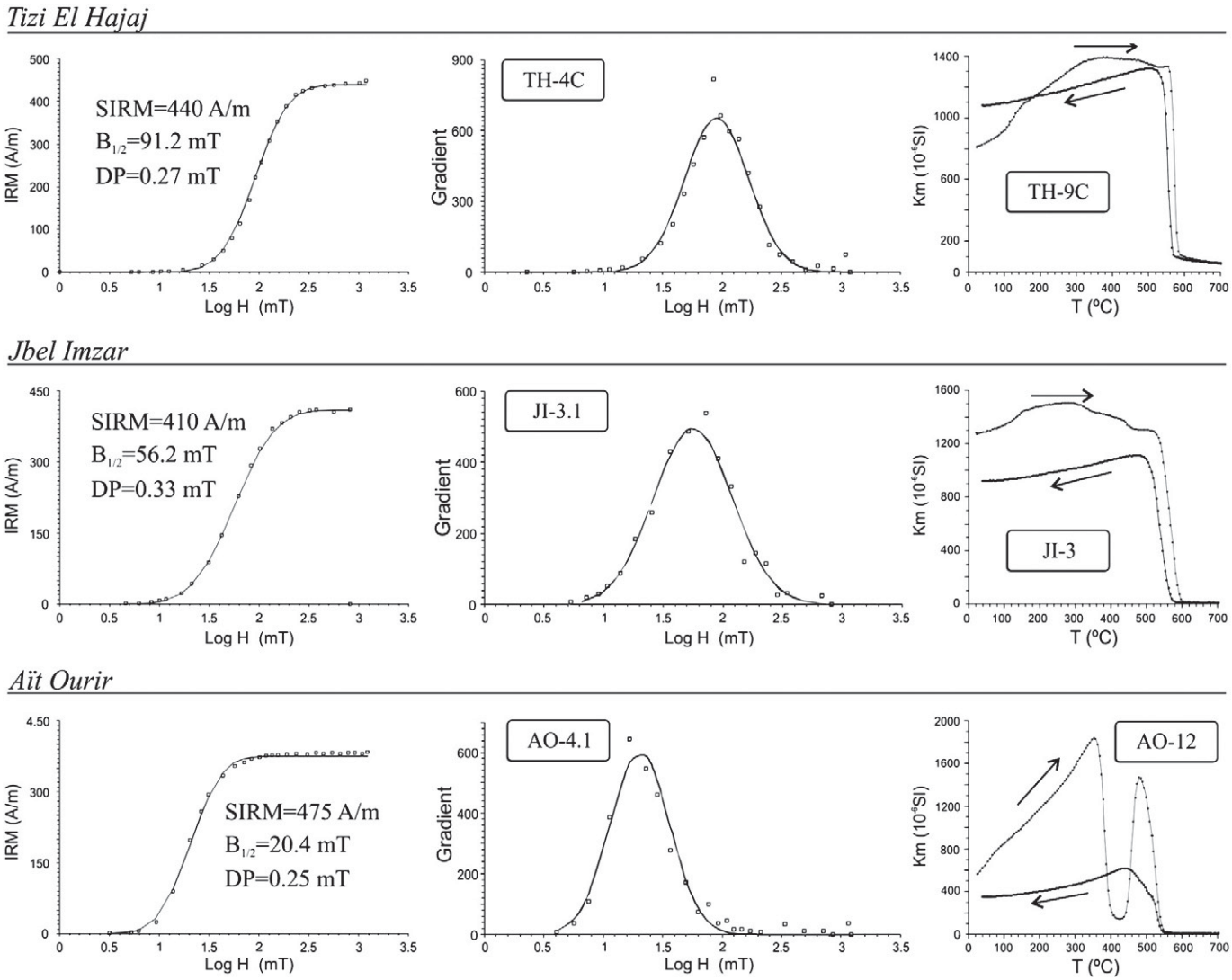


Fig. 6. Isothermal remanent magnetization (IRM) analyses treated by the cumulative log-Gaussian function (Kruiver et al., 2001) (*on the left*) and high temperature thermomagnetic curves (*on the right*).

Fisherian distribution parameters to unravel the true geomagnetic reversal nature of positive and negative polarity data sets. However, as the reversal test requires us to know the R parameter of the sample-based mean distribution and that more than one observation per polarity must be present, it was not possible to apply it to the paleomagnetic data of the Tiourjdal and Oued Lahr (Knight et al. (2004). An alternative would consist in using site-based mean directions but in both sections the negative magnetic component is present within a unique site (3 samples in Tiourjdal, 4 in Oued Lahr) thus also limiting application of the reversal test. However, after rotation by 180° of the mean ChRM computed from site-based mean directions (Table 1 in Knight et al., 2004) no overlap in their A95 area is observed indicating that they do not share the same antipodal mean magnetic component (Fisher, 1953) (Fig. 10A).

Our detailed paleomagnetic and rock magnetic study of the interbedded limestones from the Tiourjdal section rather suggest that the entire thin (~1 m) strata of limestones were affected by pervasive remagnetization processes. Sample-based mean directions are all positive (normal) and similar to the ChRM from the upper basaltic layer suggesting that the magnetic overprint mostly

Table 1
Mean magnetic component calculated from individual sample-based mean directions from the Tizi el Hajaj, Jbel Imzar and Aït Ourir sections.

	Sample-based mean directions (tilt corrected)							VGPs				
	N	Cut-off	D	I	R	K	A95	Plong	Plat	A95	K	Paleolat.
TH (Lower Unit)	36	15.5	349.2	21.7	35.739	134.1	2.1	200.5	68.2	1.7	192.6	11.2
TH (Intermediate Unit)	19	32.8	335.3	23.9	18.328	26.8	6.6	227.6	61.1	6.5	28	12.5
JI	54	27.2	338.6	23.3	52	80	3.5	62.5	238.2	3	42.5	12.1
AO	29	31.9	316.6	35.9	28.038	29.1	5.0	260.6	47.7	4.6	38.4	21.4
Paleomagnetic pole of the intermediate CAMP lava unit based on tilt corrected sample-based mean directions from all sections												
TH + JI + AO	99	31.2	333.5	27	94.781	23.2	3	241.6	60	2.6	31.4	14.3

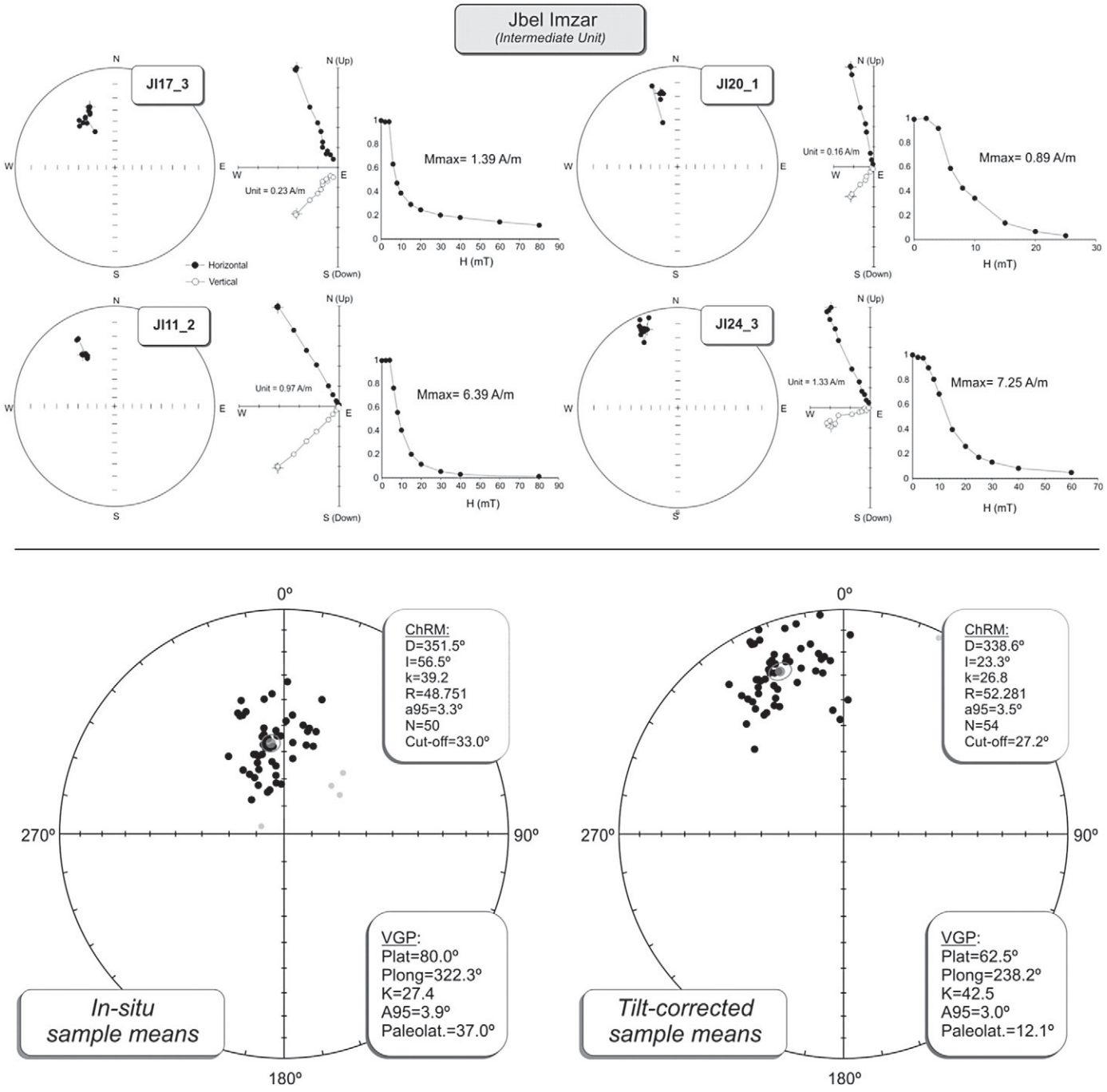


Fig. 7. Paleomagnetic results for the Jbel Imzar section.

encompassed the same time interval. Rock magnetic properties lie in the typical range of remagnetized carbonates. All samples contain a mixture of SD and MD domain ferromagnetic mineral for which position of Mrs/Ms and Hcr/Hc ratios on the Day Plot (Dunlop, 2002a,b) has a similar tendency to those of remagnetized carbonates (Channell and McCabe, 1994; Dunlop, 2002a,b; Jackson 1990; Suk et al., 1993; Tarduno and Myers, 1994; Trindade et al., 2004; Xu et al., 1998). Wasp-waisted hysteresis curves (Channell and McCabe, 1994), high Kfd (%) and ARM/IRM ratios (Jackson et al., 1992, 1993) and high values of NRM:IRMs (Fuller et al., 2002) corroborate our interpretation. Microscopic observations show rare and severely oxidized magnetites explaining the very low magnetic remanence recorded by these rocks. Pervasive enrichment in epigenetic sulfide and sulfate both as recrystallization in voids and fractures strongly suggest that these rocks were severely affected by hydrothermal alteration probably by fluid percolation from upper lavas eruption. Actually, large eruption of CAMP basalts are known to liberate huge quantity of SO_2 (producing H_2SO_4 aerosols) and CO_2 (e.g., Schaller et al., 2011; Self et al., 2006) that act as strong agents of weathering.

A second negative (reverse) component was found by Knight et al. (2004) at the base of the Intermediate Unit of the Oued Lahr section. However, as in the case of the negative magnetic component from the Tiourjdal section, when mean normal and positive magnetic components computed from site-based directions (Table 1 from Knight et al., 2004) are converted to the same hemisphere, no overlap is observed in their A95 values thus calling into question the geomagnetic origin of this antipodal component (Fig. 10A). In order to check for the presence of such a magnetic reversal, we have studied here in detail

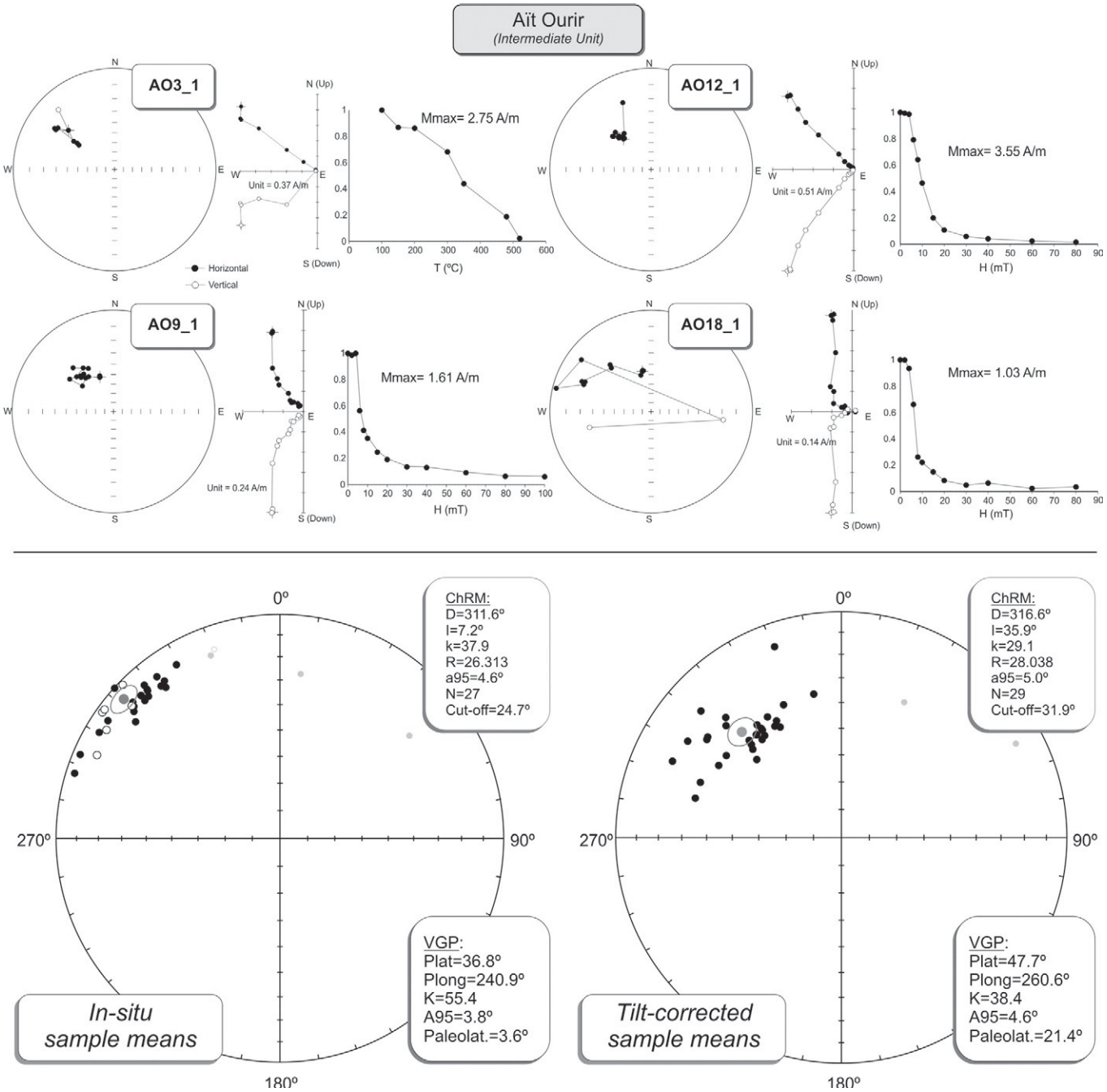


Fig. 8. Paleomagnetic results for the Aït Ourir section.

the Intermediate Unit of the CAMP lavas from three complete and distant sections. Indeed, assuming that most CAMP lavas are complete and synchronous, and this is actually suggested by a positive response in the McFadden and Lowes (1981) test when comparing the TH and JI sections, the Oued Lahr reversal should be also observed in distant contemporaneous lava flows. In each section (TH, JI and AO) site-based mean directions were calculated from each site (cylinder) sections in order to provide a detailed magnetostratigraphic scale (Fig. 9; Table X3. Result show that the entire Intermediary Unit carried a positive (normal) remanent magnetization.

In summary, we suggest that no geomagnetic reversals are present in the Moroccan CAMP lavas and that the latter mostly encompassed a large normal magnetic polarity interval correlated here to the E24n and H24n interval of the Newark and Hartford Basins respectively.

6.2. Magnetostratigraphy of the Moroccan CAMP basalts and its relationships with the Tr–J Boundary

Presence of geomagnetic reversals in the CAMP lavas is crucial to correlate the onset of the CAMP magmatism in Morocco to its trans-Atlantic continental counterparts from United States and to study its synchronism with the Tr–J crisis. However, diverse documentation of the geomagnetic polarity record across the Tr–J Boundary indicates that polarity during this time interval (late Rhaetian–early Hettangian) was mostly normal (Lucas et al., 2011 and references therein). Within this pattern, two to four very short duration reversals are expected (i.e., less than a few tens of ka; Lucas et al., 2011) that can play a crucial role to achieve precise magnetostratigraphic correlations across the Tr–J Boundary. However, such a task is limited by

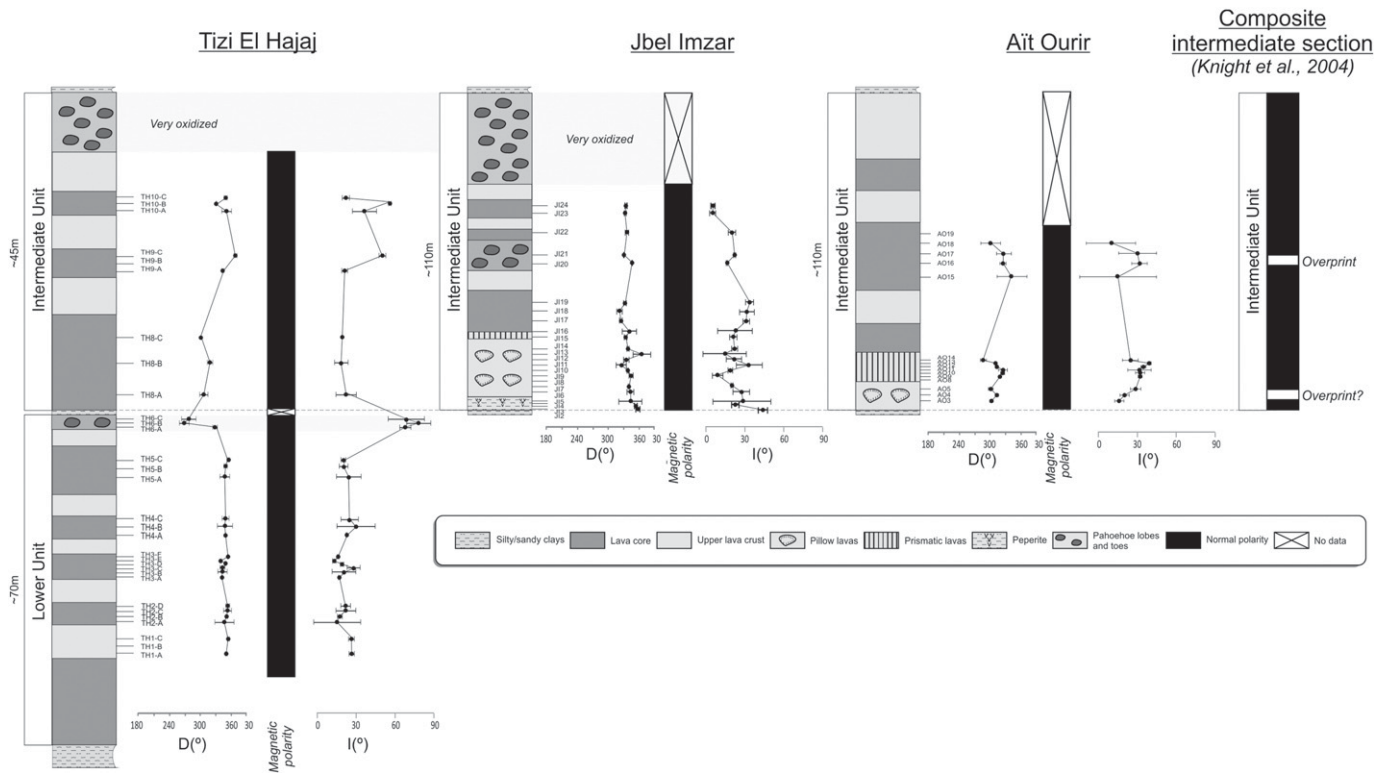


Fig. 9. Magnetostratigraphy of the Intermediate Unit of the CAMP in Morocco. All magnetic polarity intervals are positive (normal). Correlation with the composite section from Knight et al. (2004) is also shown.

depositional hiatuses, variations in sedimentation rates and remagnetization processes. In the United States, the first lavas encompassed a large interval of positive (normal) magnetic polarity, namely the chrons E24n and H24n from the Newark and Hartford Basins respectively (Kent et al., 1995; Kent and Olsen, 2008). In the same section, astronomical tuning shows that the first CAMP lavas (Orange Mountains Basalt) postdate a proposed end-Triassic continental extinction by ~20 kyr and the very short (~25 kyr) reversed magnetic polarity interval E23r by ~40 kyr (Kent and Olsen, 1999). Despite the excellent magnetostratigraphic record of the Newark Basin, correlations to CAMP lavas in Morocco still remain ambiguous (Deenen et al., 2010; Knight et al., 2004; Marzoli et al., 2004; Whiteside et al., 2007). Based on the presence of two brief reversals in the intermediate unit of CAMP lavas from the Tiourjdal and Oued Lahr sections correlated to the chron E23r from the Newark Basin, it was suggested that the CAMP in Morocco pre-dated the one in U.S. and thus could have been responsible for the Tr–J crisis (Knight et al., 2004). Conversely, Whiteside et al. (2007, 2008) rather correlated these reversals with the early Jurassic polarity reversals in the Paris basin Montcornet core (Yang et al., 1996). Correlation with the Saint Audrie's Bay is much more complicated due to the presence of several very short reverse magnetozones (SA5n2r, SA5n.3r and SA5r; Hounslow et al., 2004) that are hard to correlate with the Newark and Hartford Basins (Lucas et al., 2011, Whiteside et al., 2010). Recently, Deenen et al. (2010) found an interval of negative magnetic polarity within the infrabasaltic red siltites of the Argana Basin, Morocco, that the authors correlated with the E23r of the Newark Basin and tentatively correlated the reversals found by Knight et al. (2004) to the SA5r interval of the Saint Audrie's Bay.

Even if the relation of the Moroccan CAMP basalts with the Tr–J Boundary is beyond the scope of our study, the fact that the eruption of the Moroccan CAMP lavas encompassed a unique normal magnetozone have some implications that need to be discussed here. Beforehand, it is worth noticing that there is still no consensus for the position in the Tr–J Boundary in the marine record and comparison with the continental record is even more difficult. Secondly, bio-, chemo- and magnetostratigraphic clearly show that onset of CAMP volcanism is mostly synchronous in all circum-Atlantic basins but Moroccan volcanism may have started slightly earlier (~20 kyr) than in the Newark Basin (Deenen et al., 2010; Marzoli et al., 2011). In the Newark Basin, the Tr–J boundary was suggested to correspond to a transition in palynomorphs that coincides with a turnover in vertebrate footprints, a small iridium anomaly and a fern spore spike (Fowell et al., 1994; Olsen et al., 2002a,b; Whiteside et al., 2007). Recent research rather suggested that the boundary lies above the oldest CAMP flows in the Fundy Basin (Nova Scotia) and Morocco (Cirilli et al., 2009; Kozur and Weems, 2010; Marzoli et al., 2011). Based on $\delta^{13}C$ isotopic composition from the Newark and Hartford (U.S.A.) basins, the Saint Audrie's Bay (Nova Scotia) and the Kenecott Point (Canada), Whiteside et al. (2010) show that the end-Triassic extinction began synchronously in marine and terrestrial environments slightly before the oldest basalts in eastern North America but simultaneously with the eruption of the oldest flows in Morocco. New biostratigraphic and magnetostratigraphic data collected from the Moenave Formation, southern Colorado Plateau, Utah–Arizona, U.S.A., place the Triassic–Jurassic boundary in the middle part of the Whitmore Point Member, stratigraphically well above the occurrence of vertebrate footprints, thus suggesting that major terrestrial vertebrate extinctions preceded marine extinctions across the Triassic–Jurassic boundary and therefore were likely unrelated to the CAMP volcanism (Lucas et al., 2011).

In summary, more insights are needed to improve the correlation of the CAMP onset to the marine and terrestrial turnover across the Tr–J boundary. However, on a paleomagnetic point of view, if Moroccan CAMP lavas encompassed a unique normal magnetozone, three scenarios are basically proposed: i) the equivalent of the E23r magnetozone is represented in the Saint Audrie's Bay by the SA5r; ii) SA5n.2r or SA5n.3r are correlated to the E23r magnetozone and the

SA5r magnetozone is currently undetected within the Newark Basin and within the Moroccan CAMP lavas; and iii) the magnetostratigraphy of the Saint Audrie's Bay is suspicious and need to be revised. The first scenario better explains the absence of any reversal in the CAMP lavas compared to the magnetostratigraphy of the Saint Audrie's Bay, Newark and Hartford Basins. However, it means that the SA5n.2r and SA5n.3r should be correlated to the E22r of the Newark Basin that is more than one million years older. The second scenario was suggested by Whiteside et al. (2010, SI text) who assumed that the equivalent of reverse polarity magnetozone SA5r has been missed in the initial sampling of the Newark Basin. These authors proposed three arguments: i) most of the lower Feltville was not sampled by Kent et al. (1995); ii) the lower Feltville Formation in the Martinsville no. 1 core is highly condensed compared to elsewhere in the basin (Olsen et al., 1996) and the reverse zone could easily be omitted; and iii) they correlated SA5r to the "reversal" found in Morocco (Knight et al., 2004). However our study strongly suggests that no true geomagnetic reversals were recorded in the Moroccan CAMP lavas, thus questioning the reliability of the magnetozones isolated in the Saint Audrie's Bay (Hounslow et al., 2004). Actually, the SA5r magnetozone was isolated within very thin (0.3 to 0.6 m) stratigraphic layers and by only three specimens on two horizons.

6.3. New good quality paleomagnetic pole for CAMP basalts in Morocco

The paleomagnetic database of Northwest Africa at around 200 Ma is poorly constrained in comparison with the U.S. trans-Atlantic counterparts. Indeed only five "key" poles are considered for the interval 210–190 Ma among which three refer to the CAMP (Torsvik et al., 2008) (Table 2). Two of them have a Q factor (Van der Voo, 1990) of 3, namely the Moroccan intrusives (Bardon et al., 1973) and the Argana red beds (Morocco, Martin et al., 1978). The third one corresponds to the pole calculated by Knight et al. (2004) that unfortunately exhibits low quality statistical values (A95 = 18.5) and that differs significantly from the pole computed for U.S. counterpart (Fig. 10D). We suggest here to include a recent high quality paleomagnetic (Q = 6) pole from the Foum Zguid dyke, in Morocco, that passed a baked contact test and is thought to be mostly contemporaneous with the CAMP volcanism (Silva et al., 2006) (Table 2).

Our paleomagnetic data show that, considering individual sample-based mean directions, we can note large variations within the W° to N° quadrant, principally in declination values. This peculiar pattern is observed in the three studied sections (Figs. 5, 7 and 8) and is interpreted here to reflect the record of paleosecular variations. This hypothesis well agrees with a very short duration for the CAMP lavas (e.g., Jourdan et al., 2009; Knight et al., 2004; Marzoli et al., 2011; Olsen et al., 1996, 2003; Verati et al., 2007, Whiteside et al., 2007). In this sense, VGP calculations based on site mean directions might be erroneous and mask the original distribution of magnetic directions. Indeed, the notion of "site" in paleomagnetism is mostly related to the notion of "time" but in the case of large magmatic eruptions the acquisition time of the remanence is more controlled by the relaxation times of the magnetic carriers of the rocks, that can differ among magmatic flows, than by the moment of the eruption. Consequently, VGPs calculated using site-based mean directions can exhibit large error (A95) values of A95 and can provide an explanation for the high values of the A95 of the CAMP pole by Knight et al. (2004). Indeed, if we look at site-based mean directions of the Intermediate Unit from Tiourjdal, Oued Lahr and Aït Ourir sections (Fig. 10B), one might invoked the occurrence of diachronous volcanic pulses (i.e. distinct mean magnetic components) or error in bedding correction that is sometimes hard to distinguish in the field (Knight et al., 2004). However, when we compared these site-based mean directions with our individual sample-based means (Figs. 5, 7, 8 and 10) we clearly see that the discrepancy in declination values can be rather related to the record of paleosecular variations and that all sections share the same mean magnetic component (Fig. 10C). This interpretation is corroborated by a positive result in the McFadden and Lowes (1981) test when comparing TH (int.) and JI data sets. In contrast, the negative result obtained when comparing with the AO data set is interpreted here to be related to a limited number of samples (N = 29 with cut-off), incertitude in bedding correction as well as chemical weathering.

We thus calculated a mean paleomagnetic pole for the Middle (Intermediary Unit) CAMP in Morocco using individual VGPs calculated from tilt corrected sample-based mean directions from the Tizi El Hajaj, Jbel Imzar and Aït Ourir sections. Compilation of all data sets using the Vandamme (1994) cut-off angle gives a mean paleomagnetic pole of Plong = 241.6°, Plat = 60.0° (A95 = 2.6°, n/N = 99/106). Absence of reversal and field tests gives a Q factor of 5, with precise radiometric data from Knight et al. (2004), that can be considered as a new "key" pole for CAMP lavas in Morocco. We compared our new pole to the paleomagnetic dataset from U.S. (Kent and Olsen, 2008; Torsvik et al., 2008) after rotating North America to Northwest Africa using the Euler pole at 215 Ma (Lat = 67°, Long = 348°, Angle = 79°; ref. in Torsvik et al., 2008). After rotation, our pole better fits its U.S. counterpart (Kent and Olsen, 2008) than the previous ones.

Table 2
Selected paleomagnetic poles of Fig. 10D. Referenced in 1) Kent and Olsen (2008); 2) Torsvik et al. (2008); 3) Knight et al. (2004); and 4) Silva et al. (2006).

Key poles	Code	Plat	Plong	A95	Age (Ma)	Referenced in
North America rotated to NW Africa[a]						
Hartford SM + EB + PF (H)	K1	−69.04	60.4	2.3	201	Kent and Olsen (2008)
Moenave Wingate rotated	K2	−69.91	59.21	8.3	201	Kent and Olsen (2008)
Weston canal	K3	−69.71	60.16	2.9	207	Kent and Olsen (2008)
Martinsville	K4	−66.22	64.82	2.9	204	Kent and Olsen (2008)
CAMP U.S. (Hartford + Newark)	K5	−65.42	61.51	5	201	Kent and Olsen (2008)
Connecticut Valley volcanics	T1	−69.26	55.73	11.1	198	Torsvik et al. (2008)
Watchung basalts	T2	−67.49	51.37	6.2	201	Torsvik et al. (2008)
Hettangian Newark red beds	T3	−61.59	37.36	6	204	Torsvik et al. (2008)
Hartford, Newark basalts and volcanics	T4	−68.89	64.37	4	201	Torsvik et al. (2008)
North Mountain Basalt	T5	−76.56	62.64	10.7	200	Torsvik et al. (2008)
Passaic Fm. baked sediments	T6	−76.22	32.6	4.7	200	Torsvik et al. (2008)
Moenave Fm.	T7	−79.58	27.84	6	199	Torsvik et al. (2008)
Piedmonts dikes	T8	−83.22	38.02	7.9	199	Torsvik et al. (2008)
NW Africa poles						
MI moroccan intrusives	T9	−71	36	0	200	Torsvik et al. (2008)
Zarzaitine Fm., Algeria	T10	−70.9	55.1	2.6	206	Torsvik et al. (2008)
AR argana red beds	T11	−50.6	71.4	12	200	Torsvik et al. (2008)
CAMP — Morocco (all lavas)	K04	−73	61.3	18.5	200	Knight et al. (2004)
CAMP (Int. Unit; This study)	–	−60	62	2.6	200	This study
Foum Zguid dyke	FZ	−67.8	70.3	2.4	~200	(Silva et al., 2006)

[a] Euler pole (ref. in Torsvik et al., 2008): Lat = 67°; Long. = 348°; Angle = 79°.

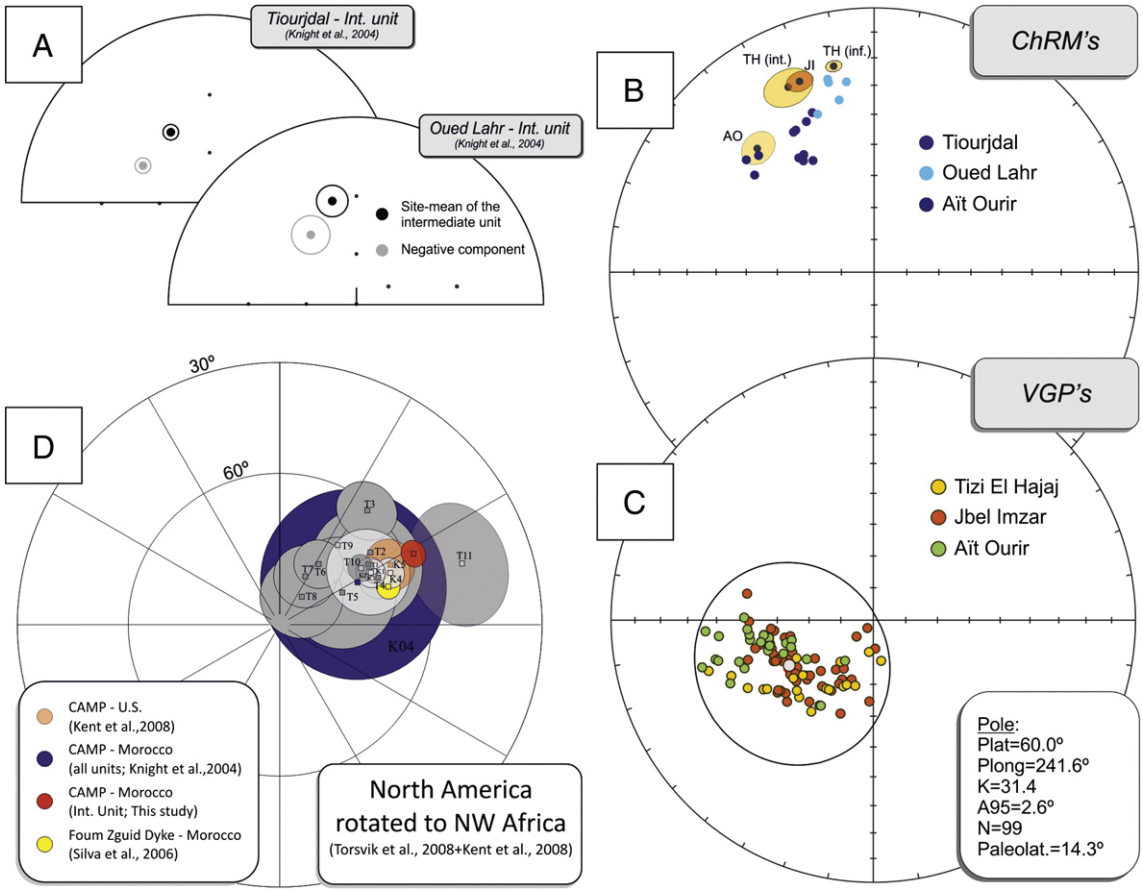


Fig. 10. A) Comparison of the positive and negative (rotated to north hemisphere) magnetic component of the Intermediate Unit in Tiourjdal and Oued Lahr sections calculated from site-based mean directions from Knight et al. (2004), absence of overlap in their A95 suggest that they do not share a common mean magnetic component; B) Comparison of mean magnetic components of the Tizi El Hajaj (Lower and Int. Units), Jbel Imzar and Aït Ourir sections calculated from sample-based mean directions with site-based mean directions from the Tiourjdal, Oued Lahr and Aït Ourir sections studied by Knight et al. (2004); C) Individual and mean VGPs calculated from sample-based mean directions (this study); D) Paleomagnetic poles from North America (Kent and Olsen, 2008; Torsvik et al., 2008) rotated to Northwest Africa using the Euler pole at 215 Ma (Lat = 67°, Long = 348°, Angle = 79°; ref. in Torsvik et al., 2008).

7. Conclusion

Our detailed paleomagnetic and rock magnetic study of the Intermediate Unit of the CAMP lavas in Morocco shows that:

1- The interbedded limestones from the Tiourjdal sections are remagnetized probably by acquisition of a Chemical Remanent Magnetization (CRM) via circulation of hot and oxidant fluids from CAMP lavas eruptions.
2- All samples from the Intermediate Unit show a positive (normal) magnetic polarity and no correlation with the reversal located at the base of the Intermediate Unit of the Oued Lahr section was found. Consequently, no geomagnetic reversals are present in the intermediate unit of the CAMP lavas in Morocco and the latter mostly encompassed a normal magnetic polarity interval.
3- It suggests that either the SA5.r magnetozone of the Saint Audrie's Bay (Nova Scotia) is not recorded in U.S. and Moroccan CAMP or the SA5.r is not so reliable and no more useful for global-scale correlations
4- Finally, we provide a new "key" pole (Q = 5) for the Middle CAMP in Morocco that is now in better agreement with its trans-Atlantic counterpart.

Supplementary materials related to this article can be found online at doi:10.1016/j.epsl.2011.07.007.

Acknowledgments

We thank Jorge Miguel Miranda (IDL, Portugal) for technical and financial supply. We also acknowledge Rachid Omira and Mohamed Khalil Bensalah for help in the field; Jean-Luc Bouchez to give access to the 2G cryogenic of the GET (Toulouse University, France) laboratory; Roberto Siqueira (GET) for technical assistance; Thierry Aigouy, Philippe de Perceval and Sophie Gouy for technical assistance in SEM analyses (GET); Cor Langereis to make its Rotepole program available; Martijn Deenen for fruitful scientific conversations; Pedro Silva (IDL, Portugal) for discussions about the Foum Zguid dyke; Marcia Ernesto and Paul Olsen for discussions about CAMP lavas. We particularly thank Anne Nédélec (GET) for internal review. We finally thanks Peter DeMenocal and anonymous reviewers for constructive reviews.

References

Aït Chayeb, E.H., Youbi, N., El-Boukhari, A., Bouabdelli, M., Amrhar, M., 1998. Le volcanisme permien et mésozoïque inférieur du bassin d'Argana (Haut-Atlas occidental, Maroc): un magmatisme intraplaque associé à l'ouverture de l'Atlantique Central. J. Afr. Earth Sci. 26, 499–519.
Bardon, C., Bossert, A., Hamzeh, R., Westphal, M., 1973. Etude paléomagnétique de formations du Trias et du Jurassique du Maroc et du Sahara. C. R. Acad. Sci. Paris Ser. 276, 2357–2360.
Bertrand, H., Dostal, J., Dupuy, C., 1982. Geochemistry of early Mesozoic tholeiites from Morocco. Earth Planet. Sci. Lett. 58, 225–239.
Channell, J.E.T., McCabe, M., 1994. Comparison of magnetic hysteresis parameters of unremagnetized and remagnetized limestones. J. Geophys. Res. 99, 4613–4623.

Cirilli, S., Marzoli, A., Tanner, L., Bertrand, H., Buratti, N., Jourdan, F., Bellieni, G., Kontak, D., Renne, P.R., 2009. Latest Triassic onset of the Central Atlantic Magmatic Province (CAMP) volcanism in the Fundy basin (Nova Scotia): new stratigraphic constraints. Earth Planet. Sci. Lett. 286, 514–525.
Cisowski, S., 1981. Interacting vs. non-interacting single domain behaviour in natural and synthetic samples. Phys. Earth Planet. Inter. 26, 56–62.
Courtillot, V.E., Renne, P.R., 2003. On the ages of flood basalt events. C. R. Geosci. 335, 113–140.
Cox, A., Doell, R.R., Dalrymple, G.B., 1964. Reversals of the earth's magnetic field. Science 144, 1537–1543.
Day, R., Fuller, M., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain size and composition dependence. Phys. Earth Planet. Inter. 13, 260–267.
Deenen, M.H.L., Ruhl, M., Bonis, N.R., Krijgsman, W., Kuerschner, W.M., Reitsma, M., van Bergen, M.J., 2010. A new chronology for the end-Triassic extinction. Earth Planet. Sci. Lett. 291, 113–125.
Dunlop, D., 2002a. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc) 1. Theoretical curves and tests using titanomagnetite data. J. Geophys. Res. 107 (B3). doi:10.1029/2001JB000486.
Dunlop, D., 2002b. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc) 2. Application to data for rocks, sediments, and soils. J. Geophys. Res. 107 (B3). doi:10.1029/2001JB000487.
El Hachimi, H., Youbi, N., Madeira, J., Bensalah, M.K., Martins, L., Mata, J., Bertrand, H., Marzoli, A., Medina, F., Munhá, J., Bellieni, J., Mahmoudi, A., Ben Abbou, M., Assafar, H., 2011. Morphology, internal architecture, and emplacement mechanisms of lava flows from the Central Atlantic Magmatic Province (CAMP) of Argana basin (Morocco). In: Van Hinsbergen, D.J.J., Buiter, S., Torsvik, T.H., Gaina, C., Webb, S. (Eds.), The Formation and Evolution of Africa: A Synopsis of 3.8 Ga of Earth History, 357, Issue 1. Special Publications, The Geological Society of London, pp. 167–193.
Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. Ser. A 217, 295–305.
Font, E., Trindade, R.I.F., Nédélec, A., 2005. Detrital remnant magnetization in hematite-bearing Neoproterozoic Puga cap dolostone, Amazon craton: a rock magnetic and SEM study. Geophys. J. Int. 163, 491–500.
Font, E., Trindade, R.I.F., Nédélec, A., 2006. Remagnetization in bituminous limestones of the Neoproterozoic Araras Group (Amazon Craton): hydrocarbon maturation, burial diagenesis or both? J. Geophys. Res. 111, B06204. doi:10.1029/2005JB004106.
Fowell, S.J., Cornet, B., Olsen, P.E., 1994. Geologically rapid Late Triassic extinctions: palynological evidence from the Newark Supergroup. In: Klein, G.D.E. (Ed.), Pangaea: Paleoclimate, Tectonics and Sedimentation During Accretion, Zenith and Break-up of a Supercontinent: Geological Society of America Special Paper, vol. 288, pp. 197–206.
Fuller, M., Kidane, T., Ali, J., 2002. AF demagnetization characteristics of NRM, compared with anhysteretic and saturation isothermal remanence: an aid in the interpretation of NRM. Phys. Chem. Earth 27, 1169–1177.
Hafid, M., Zizi, M., Bally, A.W., Ait, Salem A., 2006. Structural styles of the western onshore and offshore termination of the High Atlas, Morocco. C. R. Geosci. 338, 50–64.
Hounslow, M.W., Posen, P.E., Warrington, G., 2004. Magnetostratigraphy and biostratigraphy of the Upper Triassic and lowermost Jurassic succession, St. Audrie's Bay, UK. Palaeogeogr. Palaeoclimatol. Palaeoecol. 213, 331–358.
Jackson, M., 1990. Diagenetic sources of stable remanence in remagnetized paleozoic cratonic carbonates: a rock magnetic study. J. Geophys. Res. 95, 2753–2762.
Jackson, M., Worm, H., 2001. Anomalous unblocking temperatures, viscosity and frequency-dependent susceptibility in the chemically-remagnetized Trenton limestone. Phys. Earth Planet. Inter. 126, 27–42.
Jackson, M., Sun, W.W., Craddock, J.P., 1992. The rock magnetic fingerprint of chemical remagnetization in midcontinental paleozoic carbonates. Geophys. Res. Lett. 19, 781–784.
Jackson, M., Rochette, P., Filon, G., Banerjee, S., Marvin, J., 1993. Rock magnetism of remagnetized Paleozoic carbonates: low-temperature behaviour and susceptibility characteristics. J. Geophys. Res. 98, 6217–6225.
Johnson, H.P., Lowrie, W., Kent, D.V., 1975. Stability of anhysteretic remanent magnetization in fine and coarse magnetite and maghemite. Geophys. J. R. Astron. Soc. 41, 1–10.
Jourdan, F., Marzoli, A., Bertrand, H., Cirilli, S., Tanner, L., Kontak, D.J., McHone, G., Renne, P.R., Bellieni, G., 2009. $^{40}Ar/^{39}Ar$ ages of CAMP in North America: implications for the Triassic–Jurassic boundary and the ^{40}K decay constant bias. Lithos 110, 167–180.
Katz, B., Elmore, R.D., Cogoini, M., Engel, M.H., Ferry, S., 2000. Associations between burial diagenesis of smectite, chemical remagnetization, and magnetite authigenesis in the Vocontian trough, SE France. J. Geophys. Res. 105, 851–868.
Kent, D.V., Olsen, P.E., 1999. Astronomically tuned geomagnetic polarity time scale for the Late Triassic. J. Geophys. Res. 104, 12831–12841.
Kent, D.V., Olsen, P.E., 2008. Early Jurassic magnetostratigraphy and paleolatitudes from the Hartford continental rift basin (eastern North America): testing for polarity bias and abrupt polar wander in association with the central Atlantic magmatic province. J. Geophys. Res. 113, 1–24.
Kent, D.V., Olsen, P.E., Witte, W.K., 1995. Late Triassic–earliest Jurassic geomagnetic polarity sequence and paleolatitudes from drill cores in the Newark rift basin, eastern North America. J. Geophys. Res. 100, 14,965–14,998. doi:10.1029/95JB01054.
Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.
Knight, K.B., Nomade, S., Renne, P.R., Marzoli, A., Bertrand, H., Youbi, N., 2004. The Central Atlantic Magmatic Province at the Triassic–Jurassic boundary: paleomagnetic and $^{40}Ar/^{39}Ar$ evidence from Morocco for brief, episodic volcanism. Earth Planet. Sci. Lett. 228, 143–160.
Kozur, H.W., Weems, R.E., 2010. The biostratigraphic importance of conchostracans in the continental Triassic of the northern hemisphere. In: Lucas, S.G. (Ed.), The Triassic Timescale. Geological Society, 334. Special Publications, London, pp. 315–417.
Kruiver, P.P., Dekkers, M.J., Heslop, D., 2001. Quantification of magnetic coercivity components by the analysis of acquisition curves of isothermal remanent magnetization. Earth planet. Sci. Lett. 189, 269–276.
Kuiper, K.F., Deino, A., Hilgen, F.J., Krijgsman, W., Renne, P.R., Wijbrans, J.R., 2008. Synchronizing rock clocks of Earth history. Nature 20, 500–504.
Lucas, S.G., Tanner, L.H., 2007. The nonmarine Triassic–Jurassic boundary in the Newark Supergroup of eastern North America. Earth Sci. Rev. 84, 1–20.
Lucas, S.G., Tanner, L.H., Donohoo-Hurley, L.L., Geissman, J.W., Kozur, H.W., Heckert, A.B., Weems, R.E., 2011. Position of the Triassic–Jurassic boundary and timing of the end-Triassic extinctions on land: data from the Moenave Formation on the southern Colorado Plateau, USA. Palaeogeogr. Palaeoclimatol. Palaeoecol. 302, 194–205.
Martin, D.L., Nairn, A.E.M., Noltimier, H.C., Petty, M.H., Schmidt, T.J., 1978. Paleozoic and Mesozoic paleomagnetic results from Morocco. Tectonophysics 44, 91–114.
Marzoli, A., Renne, P.R., Piccirillo, E.M., Ernesto, M., Bellieni, G., De Min, A., 1999. Extensive 200-million-year-old continental flood basalts of the Central Atlantic Magmatic Province. Science 284, 616–618.
Marzoli, A., Bertrand, H., Knight, K.B., Cirilli, S., Buratti, N., Verati, C., Nomade, S., Renne, P.R., Youbi, N., Martini, R., Allenbach, K., Neuwerth, R., Rapaille, C., Zaninetti, L., Bellieni, G., 2004. Synchrony of the Central Atlantic magmatic province and the Triassic–Jurassic boundary climatic and biotic crisis. Geology 32, 973–976.
Marzoli, A., Bertrand, H., Chiaradia, M., Fontignie, D., Youbi, N., Bellieni, G., 2006. The genesis of CAMP basalts (Morocco) from enriched lithosphere to late asthenosphere mantle sources. Geochim. Cosmochim. Acta 70, A396.
Marzoli, A., Bertrand, H., Knight, K., Cirilli, S., Nomade, S., Renne, P.R., Vérati, C., Youbi, N., Martini, R., Bellieni, G., 2008. Synchrony between the Central Atlantic magmatic province and the Triassic–Jurassic mass-extinction event? Comment. Palaeogeogr. Palaeoclimatol. Palaeoecol. 262, 189–193.
Marzoli, A., Jourdan, F., Puffer, J.H., Cuppone, T., Tanner, H., Weems, R.E., Bertrand, H., Cirilli, S., Bellieni, G., De Min, A., 2011. Timing and duration of the Central Atlantic magmatic province in the Newark and Culpeper basins, eastern U.S.A. Lithos 122, 175–188.
Matuyama, M., 1929. On the direction of magnetization of basalt in Japan, Tyosen and Manchuria. Proc. Imp. Acad. Japan 5, 203–205.
McCabe, C., Elmore, D.R., 1989. The occurrence and origin of late paleozoic remagnetization in the sedimentary rocks of north America. Rev. Geophys. 27, 471–494.
McFadden, P.L., Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J. R. Astron. Soc. 67–1, 19–33.
McFadden, P., McElhinny, M.W., 1990. Classification of the reversal test in paleomagnetism. Geophys. J. Int. 103, 725–729.
Mercanton, P.I., 1926. Inversion de l'inclinaison magnétique terrestres aux âges géologiques. Terr. Magn. Atmos. Elec. 31, 187–190.
Morgan, P.J., Reston, T.J., Ranero, C.R., 2004. Contemporaneous mass extinctions, continental flood basalts, and 'impact signals': are mantle plume-induced lithospheric gas explosions the causal link? Earth Planet. Sci. Lett. 217, 263–284.
Newell, N.D., 1967. Revolutions in the history of life. Geol. Soc. Am. Spec. Pap. 89, 63–91.
Olsen, P.E., Shubin, N.H., Anders, P.E., 1987. New Early Jurassic tetrapod assemblages constrain Triassic–Jurassic tetrapod extinction event. Science 237, 1025–1029.
Olsen, P.E., Schlische, R.W., Fedosh, M.S., 1996. 580 Ky duration of the early Jurassic flood basalt event in eastern North America estimated using Milankovitch cyclostratigraphy. Bull. Mus. North. Arizona 60, 11–22.
Olsen, P.E., Kent, D.V., Sues, H.-D., Koeberl, C., Huber, H., Montanari, A., Rainforth, E.C., Fowell, S.J., Szajna, M.J., Hartline, B.W., 2002a. Ascent of dinosaurs linked to an iridium anomaly at the Triassic–Jurassic boundary. Science 296, 1305–1307.
Olsen, P.E., Koeberl, C., Huber, H., Montanari, A., Fowell, S.J., Et-Touhami, M., Kent, D.V., 2002b. Continental Triassic–Jurassic boundary in central Pangea: recent progress and discussion of an Ir anomaly. In: Koerberl, C., MacLeod, K.G. (Eds.), Catastrophic Events and Mass Extinctions: Impacts and Beyond: Boulder, Colorado: Geological Society of America Special Paper, vol. 356, pp. 502–522.
Olsen, P.E., Kent, D.V., Et-Touhami, M., Puffer, J., 2003. Cyclo-, magneto-, and biostratigraphic constraints on the duration of the CAMP event and its relationship to the Triassic–Jurassic boundary. AGU Geophys. Mon. . 7–32 pp.
Raup, D.M., Sepkoski Jr., J.J., 1982. Mass extinctions in the marine fossil record. Science 215, 1501–1503.
Renne, P.R., Mundil, R., Balco, G., Min, K., Ludwig, K.R., 2010. Joint determination of ^{40}K decay constants and $^{40}Ar^*/^{40}K$ for the Fish Canyon sanidine standard, and improved accuracy for $^{40}Ar/^{39}Ar$ geochronology. Geochim. Cosmochim. Acta 74, 5349–5367.
Robertson, D.J., France, D.E., 1994. Discrimination of remanence-carrying minerals in mixtures using isothermal remanent magnetisation acquisition curves. Phys. Earth Planet. Inter. 82, 223–234.
Schaller, M.n., Wright, J.D., Kent, D.V., 2011. Atmospheric pCO_2 perturbations associated with the Central Atlantic Magmatic Province. Science. doi:10.1126/science.1199011.
Schaltegger, U., Guex, J., Bartolini, A., Schoene, B., Ovtcharova, M., 2008. Precise U–Pb age constraints for end-Triassic mass extinction, its correlation to volcanism and Hettangian post-extinction recovery. Earth Planet. Sci. Lett. 267, 266–275.
Self, S., Thordarson, T.H., Keszthelyi, L., 1997. Emplacement of continental flood basalt lava flows. In: Mahoney, J.J., Coffin, M.F. (Eds.), Large Igneous Provinces: Continental, Oceanic, and Planetary Flood Volcanism: AGU Geophys. Monogr. Ser., 100, pp. 381–410.
Self, S., Keszthelyi, L., Thordarson, T.H., 1998. The importance of pahoehoe. Ann. Rev. Earth Planet. Sci. 26, 81–110.

Self, S., Widdowson, M., Thordarson, T., Jay, A.E., 2006. Volatile fluxes during flood basalt eruptions and potential effects on the global environment: a Deccan perspective. Earth Planet. Sci. Lett. 248, 517–531.

Silva, P.M.F., Henry, B., Marques, F.O., Madureira, P., Miranda, J.M., 2006. Paleomagnetic study of the Great Foum Zguid dyke (southern Morocco): a positive contact test related to metasomatic processes. Geophys. Res. Lett. 33, L21301. doi:10.1029/2006GL027498.

Suk, D., Van der Voo, R., Peacor, D.R., 1993. Origin of magnetite responsible for remagnetization of early Palaeozoic limestones of New York State. J. Geophys. Res. 98, 419–434.

Tanner, L.H., Kyte, F.T., Walker, A.E., 2008. Multiple Ir anomalies in uppermost Triassic to Jurassic-age strata of the Blomidon Formation, Fundy basin, eastern Canada. Earth Planet. Sci. Lett. 274, 103–111.

Tarduno, J.A., Myers, M., 1994. A primary magnetization fingerprint from the Cretaceous Laytonville calcáreos: further evidence for rapid oceanic plate velocities. J. Geophys. Res. 99, 21691–21703.

Torsvik, T.H., Müller, R.D., Van der Voo, R., Steinberger, B., Gaina, C., 2008. Global plate motion frames: toward a unified model. Rev. Geophys. 46, 1–44 RG3004.

Trindade, R.I.F., D'Agrella-Filho, M.S., Babinski, M., Font, E., Brito Neves, B.B., 2004. Paleomagnetism and geochronology of the Bebedouro cap carbonate: evidence for continental-scale Cambrian remagnetization in the São Francisco craton, Brazil. Precambrian Res. 128, 83–103.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184 (1), 1–9.

Vandamme, D., 1994. A new method to determine paleosecular variation. Phys. Earth Planet. Inter. 85, 131–142.

Verati, C., Rapaille, C., Feraud, G., Marzoli, A., Bertrand, H., Youbi, N., 2007. $^{40}Ar/^{39}Ar$ ages and duration of the Central Atlantic Magmatic Province volcanism in Morocco and Portugal and its relation to the Triassic–Jurassic boundary. Palaeogeogr. Palaeoclimatol. Palaeoecol. 244, 308–325.

Ward, P.D., Haggart, J.W., Carter, E.S., Wilbur, D., Tipper, H.W., Evans, T., 2001. Sudden productivity collapse associated with the Triassic–Jurassic boundary mass extinction. Science 292, 1148–1151.

Weil, A.B., Van der Voo, R., 2002. Insights into the mechanism for orogen-related carbonate remagnetization from growth of authigenic Fe-oxide: a scanning electron microscopy and rock magnetic study of Devonian carbonates from northern Spain. J. Geophys. Res. 107. doi:10.1029/2001JB000200.

Whiteside, J.H., Olsen, P.E., Kent, D.V., Fowell, S.J., Et-Touhami, M., 2007. Synchrony between the Central Atlantic magmatic province and the Triassic–Jurassic massextinction event? Palaeogeogr. Palaeoclimatol. Palaeoecol. 244, 345–367.

Whiteside, J.H., Olsen, P.E., Kent, D.V., Fowell, S.J., Et-Touhami, M., 2008. Synchrony between the Central Atlantic magmatic province and the Triassic–Jurassic massextinction event? Reply to Marzoli et al. Palaeogeogr. Palaeoclimatol. Palaeoecol. 262, 194–198.

Whiteside, J.H., Olsen, P.E., Eglinton, T., Brookfield, M.E., Sambrotto, R.N., 2010. Compound-specific carbon isotopes from Earth's largest flood basalt eruptions directly linked to the end-Triassic mass extinction. . doi:10.1073/pnas.1001706107.

Xu, W., Van der Voo, R., Peacor, D.R., 1998. Electron microscopic and rock magnetic study of remagnetized Leadville carbonates, central Colorado. Tectonophysics 296, 333–362.

Yang, Z.Y., Moreau, M.G., Bucher, H., Dommergues, J.L., Trouiller, A., 1996. Hettangian and Sinemurian magnetostratigraphy from Paris Basin. J. Geophys. Res. 101, 8025–8042.

Youbi, N., Martins, L.T., Munha, J.M., Ibouh, H., Madeira, J., Chayeb, A., El Boukhari, A., 2003. The Late Triassic–Early Jurassic volcanism of Morocco and Portugal in the geodynamic framework of the opening of the Central Atlantic ocean. In: Hames, W.E., McHone, J.G., Renne, P.R., Ruppel, C. (Eds.), The Central Atlantic Magmatic Province: Insights from Fragments of Pangea. : Geophysical Monograph, 136. American Geophysical Union, Washington, pp. 179–207.

Zwing, A., Matzka, J., Bachtadse, V., Soffel, H.C., 2005. Rock magnetic properties of remagnetized Palaeozoic clastic and carbonate rocks from the NE Rhenish massif, Germany. Geophys. J. Int. 160, 477–486.